PMCNet: A Physical Model-Constrained Network for Magnetic Particle Imaging Reconstruction

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Magnetic particle imaging (MPI) quantitatively visualizes the spatial distribution of magnetic nanoparticles by reconstructing volumetric image of their nonlinear response signals. Deep learning techniques supported by large amounts of training data has have been used in MPI reconstruction. However, the acquisition of MPI data requires a long time of scanning and preprocessing. This makes it impractical and unavailable to acquire a sufficient amount of data to construct a high-quality training dataset. In this work, we propose a physical model-constrained network (PMCNet) for MPI reconstruction in which the physical model constraints are incorporated into deep learning networks to overcome the limitation of dependence on training data. The PMCNet evolves through the constraints of the physical model rather than being trained with paired data. In the method, the neural networks are run to generate a spatial distribution of magnetic nanoparticles, and then a physical model is used to calculate the magnetic response signals generated from this spatial distribution. The parameters of the neural networks are optimized by minimizing the error between the measured and calculated magnetic response signals. The performance of the PMCNet is evaluated with a series of simulations. The results collectively demonstrate that the PMCNet has high accuracy, high robustness, and good generalizability, facilitating high-resolution and multi-color MPI. Although PMCNet demonstrates superior performance in controlled simulations, its evaluation on the OpenMPI public dataset reveals a performance gap compared to results from synthetic data. This discrepancy primarily stems from the inherent challenge of simplified forward model in fully capturing the complex physical non-idealities present in real-world hardware environments.

***Index Terms*—Magnetic particle imaging (MPI), physical model, image reconstruction, deep learning.**

I. Introduction

MAGNETIC particle imaging (MPI) is a new quantitative and functional medical imaging technology, which visualizes magnetic nanoparticles (MNPs) in space through the nonlinear magnetization[1]. It reconstructs the spatial distribution of particles by using the voltage signal induced from the receiving coil. System matrix-based reconstruction [2-4] and X-space reconstruction [5-6] are two primarily used reconstruction algorithms for MPI. The former converts the reconstruction into a problem of solving an ill-posed linear equation. It describes the relationship between the distribution of MNPs and measured voltage signals which is established via the system matrix [7]. However, the system matrix-based (SM) reconstruction needs to construct the system matrix in advance, which is a very time-consuming process. Although the system matrix can be calculated via modeling, the gap between the model and the real scanning will affect the final imaging results [8]. The latter X-space reconstruction can reflect the distribution and concentration of MNPs from time-domain signal in real-time [9]. The reconstructed image is the result of the convolution of MNPs concentration and point spread function. This method may produce artifacts in the reconstructed image and cause a decrease in resolution.

The rapid development of deep learning has inspired new approaches for MPI reconstruction. Pure data-driven methods train networks to recover images directly from frequency- or time-domain data [10–13]. Incorporating deep learning priors into computational imaging models has further improved system matrix (SM) image quality [14–16]. In these approaches, deep neural networks often serve as regularizers to solve linear reconstruction equations [17–18], which still require time-consuming system matrix calibration. Baltruschat et al. [14] first introduced a super-resolution convolutional neural network (CNN), originally designed for natural images, to address this issue. Later, Shi et al. [19] proposed a progressive transformer network that integrates coil and frequency information. However, most deep learning-based MPI reconstruction methods still rely on training data or system matrices. Inspired by recent advances in 3D optical imaging [20], model-constrained untrained networks have been proposed to reconstruct images without pre-training, effectively bypassing these limitations.

In contrast to existing deep learning-based MPI reconstruction methods that rely on large training datasets or pre-calibrated system matrices, our previous work [21] introduced a data-free approach that optimizes network parameters directly under physical model constraints. Building upon this idea, we propose PMCNet (Physical Model-Constrained Network), a novel framework that embeds the MPI physical model into the network architecture itself. This integration enables image reconstruction without any training data or system matrix calibration. Simulation results demonstrate that PMCNet achieves higher accuracy, robustness, and generalizability than the conventional X-space method. Moreover, the framework naturally extends to relaxation and multicolor MPI, further highlighting its versatility and potential.

II. Methods

A. *PMCNet framework*

The diagram of the PMCNet-based reconstruction framework is shown in Figure 1. The framework mainly consists of three parts: the reconstruction network, the physical model, and the loss function. The neural network is used to generate the reconstructed results, which is the spatial distribution of MNPs concentration. The physical model describes the forward process of MPI mathematically, which can calculate the voltage signal corresponding to the distribution of current MNPs' concentration. The loss function is constructed by calculating the average absolute error between the voltage signal calculated by the physical model and the measured signal. By optimizing the network parameters, the final reconstruction result can be obtained. In the simulation, u_{meas} and u use the same physical model for calculation, but in the experiment, u_{meas} will be obtained through measurement.

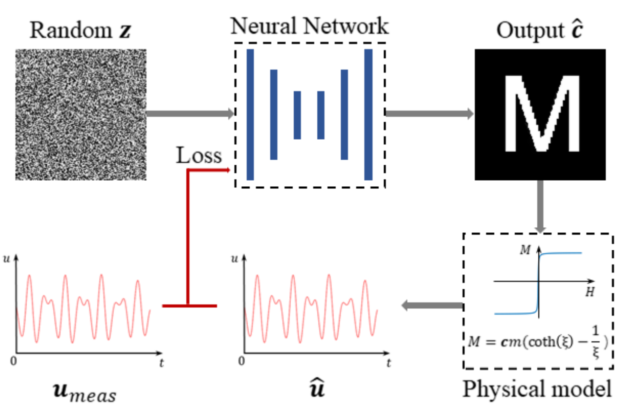

Fig. 1. PMCNet framework. The network predicts a concentration distribution *c*, which is then used to generate a corresponding signal $\hat{u}$, through the physical model (e.g., the Langevin function). This modeled signal is compared with the measured signal u_{meas} and the absolute difference between them is minimized iteratively during optimization.

B. *MPI physical model: Langevin model*

First, we use the ideal model in MPI. Under the assumption of this model, the relationship between the time-domain induced voltage $u(t)$ in receiving coil and MNPs' concentration $c(\boldsymbol{r})$ can be [22]:

$$u(t) = -\mu_0 \int_{\Omega} c(\boldsymbol{r}) p(\boldsymbol{r}) \frac{\partial}{\partial t} \boldsymbol{M}(\boldsymbol{r},t) d\boldsymbol{r} \tag{1}$$

where μ_0 is the permeability of a vacuum, $p(\boldsymbol{r})$ is the sensitivity profile of the recording coil, and $\boldsymbol{M}(\boldsymbol{r},t)$ denotes the unit magnetization of the MNPs. The magnetization model of the magnetization response of MNPs $\boldsymbol{M}(\boldsymbol{r},t)$ is characterized by the Langevin function [21]:

$$\boldsymbol{M}(\boldsymbol{r},t) = \boldsymbol{e_H} cm(\coth(\beta H) - \frac{1}{\beta H}) A/m \tag{2}$$

with

$$\beta = \frac{\mu_0 m}{k_B T} \tag{3}$$

where $\boldsymbol{e_H}$ denotes the direction of the magnetic field strength, c is the concentration of MNPs, m is the magnetic moment, $\coth(\bullet)$ denotes the hyperbolic cotangent function, H is the magnetic field strength, μ_0 is the magnetic permeability, k_B is Boltzmann constant, and T is temperature.

Equations (1)-(3) denote ideal model (instantaneous thermodynamic equilibrium) of MPI, which can be used as the physical model to obtain the corresponding voltage signal when the distribution of magnetic particle concentration is known. In this study, this ideal model was selected to validate the accuracy, spatial resolution, and robustness of the PMCNet framework.

C. *MPI physical model: Debye relaxation model*

The actual scanning process of MPI is usually nonadiabatic. In the non-adiabatic environment, the magnetization dynamics of MNPs would be affected. Magnetic nanoparticles cannot immediately respond to changes in the external field; instead, there is a finite response time. This phenomenon is referred as relaxation effect. MNPs being of different types or in different microenvironments typically exhibit varying degrees of relaxation effects. By distinguishing the strength of relaxation, the particle types or environmental parameters can be estimated, enabling multi-color MPI [23]. Currently, this phenomenon can be described by the first-order Debye relaxation model [24]:

$$\frac{d\boldsymbol{M_D}(\boldsymbol{r},t)}{dt} = \frac{\boldsymbol{M_D}(\boldsymbol{r},t) - \boldsymbol{M}(\boldsymbol{r},t)}{\tau} \tag{4}$$

where $\boldsymbol{M_D}(\boldsymbol{r},t)$ denotes the Debye model that describes non-adiabatic magnetization, τ denotes a relaxation time constant. By solving this differential equation, the temporal convolution formulation can be obtained:

$$\begin{aligned} \boldsymbol{M}_D(\boldsymbol{r},t) &= \boldsymbol{M}(\boldsymbol{r},t) * \frac{1}{\tau}\exp(-\frac{t}{\tau})u(t) \\ &= \boldsymbol{M}(\boldsymbol{r},t) * r(t) \end{aligned} \tag{5}$$

where $u(t)$ denotes the Heaviside function. Non-adiabatic

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magnetization can be described as the temporal convolution of the adiabatic magnetization and convolution kernel $r(t)$. This Debye relaxation model of (4)-(5) was employed in the subsequent relaxation and multicolor experiments.

For both the Debye model and Langevin model, the procedure is the same, but the models used for signal calculation are different. The Debye model uses the Debye function, while the Langevin model uses the Langevin function to calculate the signals. The reconstruction goal is to minimize the difference between the simulated signal and the measured signal.

D. *Structure of neural network*

The neural network adopts a structure similar to U-Net to obtain the spatial distribution of MNPs concentration. The network structure is depicted in Fig. 2, which includes an encoder and a decoder. The network consists of four down-sampling modules and four up-sampling modules. The down-sampling module uses the Max pooling layer to achieve the down-sampling operation of halving the image size, while the up-sampling restores the original size of the image through the transposed convolutional layer. Since this network doesn't need to learn the mapping from voltage signals to the distribution of MNPs concentration, a random matrix was used as the network input. This process can be represented as:

$$\hat{\boldsymbol{c}} = \varphi_\theta(\mathbf{z}) \tag{6}$$

where φ_θ denotes the neural network used for reconstructing the MNPs distribution from the input, θ represents the network parameters, z represents the input of the network, and $\hat{\boldsymbol{c}}$ represents the reconstructed MNPs distribution and denotes the output of the network. To avoid the impact of shallow noise on the reconstruction results, different from UNet, the shallow skip connection structure is eliminated in our design.

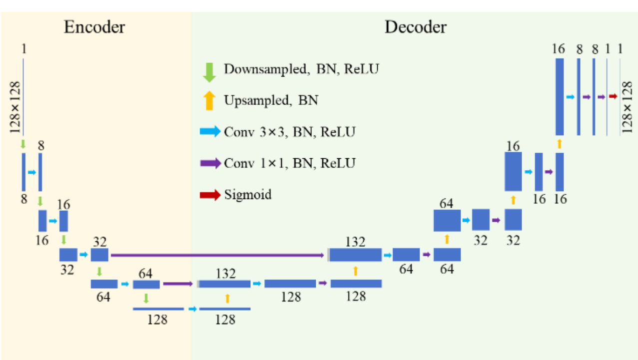

Fig. 2. Structure of neural networks for MNPs reconstruction. The light-yellow part is the Encoder, and the light green part is the Decoder. The green and yellow arrows respectively represent the downsampling module and the upsampling module. The blue and purple ones are two convolutional modules with different kernel sizes.

E. *Loss function*

In theory, as the reconstruction quality of the network improves, the voltage signal obtained from the physical model should also become increasingly closer to the voltage signal measured by the MPI system. Therefore, we update the network parameters by minimizing the difference between these two voltages:

$$\theta = \arg\min_\theta \left\| P(\varphi_\theta(\mathbf{z})) - \boldsymbol{u}_{meas} \right\|_1^1 \tag{7}$$

where $P(\cdot)$ represents the physical model. To mitigate the influence of noise in the received signal on the reconstruction results, we opt to update the network parameters using mean absolute error instead of mean squared error. The network parameters are optimized using Adam. The final loss function can be expressed as:

$$Loss = \left\| P(\varphi_\theta(\mathbf{z})) - \boldsymbol{u}_{meas} \right\|_1^1 \tag{8}$$

III. Experimental Setting

In the simulation experiment of multi-color MPI, the non-adiabatic model was employed as the forward model; otherwise, the adiabatic model was selected.

A. *Simulation data acquisition*

We first employed a dot phantom to evaluate the spatially discriminative ability of the PMCNet framework by drawing a point spread function (PSF). Additionally, three different types of phantoms, including letter, concentration, and vessel phantoms, as illustrated in Figure 3, were designed to explore the accuracy and robustness of the PMCNet framework. Line-pair phantoms with different spacings were designed to define the spatial resolution of the proposed approach. Furthermore, we designed experiments to validate the relaxation data and multi-color MPI reconstruction of PMCNet.

The phantoms consisted of 128×128 pixels, with each pixel of 0.2 mm×0.2 mm, and the diameter of MNPs is set to 30 nm. The reception signals under adiabatic conditions were simulated using the Langevin function. The gradient field was set to be 2 T/m. The driving field comprised of a 24.5 kHz sinusoidal waves along the x-axis and a 25.25 kHz sinusoidal waves along the y-axis with an amplitude of 28 mT, generating a Cartesian scanning trajectory. In the robustness experiments, various levels of Gaussian white noise were added to the reception signals. The noise is added to the signal, rather than disturbing the model. In the relaxation and multi-color MPI experiments, the Debye relaxation kernel was employed to simulate relaxation phenomena during scanning.

B. *Physical model settings*

Considering that the parameters of the MPI system are known in the actual experiments, we set the parameters of the forward model of PMCNet to be consistent with the simulation process. However, the forward model utilized the output of the reconstruction network as the concentration distribution of the MNPs. When reconstructing data containing relaxation effects, we did not provide the magnitude of the relaxation time constant directly, but instead estimated it through the gradient descent algorithm applied to the loss function.

The neural network was implemented on the Python 3.8.0

platform using Pytorch of version 1.11.0. We used the Adam optimizer to optimize the network parameters with a learning rate of 10^{-3}. 20000 epochs were used for each data to ensure a good reconstructed image. Network training is performed on an NVIDIA TITAN GTX GPU.

C. Evaluation Metrics

The peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) are introduced to evaluate the quality of reconstructed images [23]. PSNR calculates the ratio between the maximum possible signal power and the error power in an image. Thus, it was employed to indicate the impact of noise on the quality of the reconstructed image.

$$PSNR(I,\hat{I}) = 20\log_{10}\frac{\sqrt{N}\,\|I\|_\infty}{\|I-\hat{I}\|_2} \tag{9}$$

where I is the ground truth, $\hat{I}$ is the reconstructed image, and N is the number of pixels. The SSIM is a metric based on perceivable visual quality that better reflects human evaluation of the quality of the reconstructed images:

$$SSIM(I,\hat{I}) = \frac{(2\mu_I\mu_{\hat{I}}+c_1)(2\sigma_{I\hat{I}}+c_2)}{(\mu_I^2+\mu_{\hat{I}}^2+c_1)(\sigma_I^2+\sigma_{\hat{I}}^2+c_2)} \tag{10}$$

where μ_I, $\mu_{\hat{I}}$ are the means of the reconstruction result and ground truth; $c_1=(k_1L)^2$, $c_2=(k_2L)^2$, $k_1=0.01$ and $k_2=0.03$, L represents the range of pixels. σ_I^2, $\sigma_{\hat{I}}^2$ represent the variances and $\sigma_{I\hat{I}}$ is the covariance between the reconstruction result and the ground truth.

D. Experiment design

To verify the effectiveness of the PMCNet framework, we conducted several simulation-based comparative experiments, in which the results of the PMCNet were compared with both the X-Space and SM methods.

Considering the external disturbances present in the actual scanning process, we added Gaussian noise of different degrees to the simulated voltages of the simulation experiments to explore the robustness of the PMCNet framework. In the first half of the simulation experiments, we conducted experiments based on the adiabatic assumption. However, to simulate the relaxation effect, we used the Debye relaxation model as the physical model of the PMCNet framework. We explored the effectiveness of the PMCNet framework for reconstructing non-adiabatic data and further investigated the possibility of its application in multi-color MPI.

IV. Results

A. Accuracy evaluation

First, we evaluated the performance of the PMCNet with a dot phantom in the ideal scenario without fundamental frequency loss by comparing with the ground truth (Figure 3(a)). It can be seen that the X-Space reconstruction method is affected by the PSF, resulting in radial artifacts around the dot phantom, which will limit its spatial resolution. However, our PMCNet method avoids artifacts through a special reconstruction mechanism of fusing deep learning networks with physical models. From the cross-sectional view, we can see that there are no radial artifacts in PMCNet results, which are well consistent with the ground truth and SM method. This also reflects the potential of PMCNet in improving the spatial resolution of MPI reconstructions.

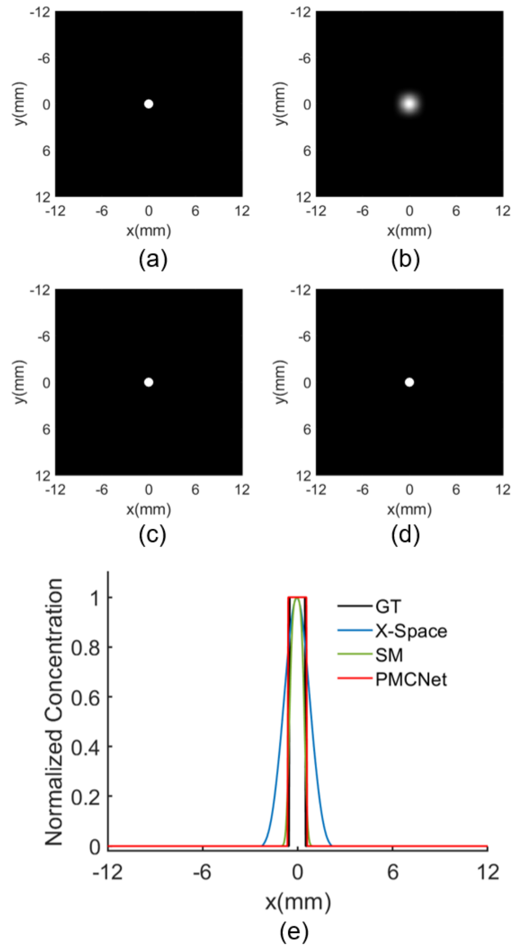

Fig. 3. Reconstructed results of a simulated dot phantom. (a) The simulated dot phantom. (b) The reconstructed result of the X-Space method. (c) The reconstructed result of the SM method. (d) The reconstructed result of the PMCNet framework. (e) Intensity profiles of reconstructed results on a line extracted through the center of the dot phantom (at y=0 mm).

Next, we validated the quality of the reconstructed results in different scenarios, including three designed phantoms (Fig. 4). These phantoms correspond to more complex reconstruction scenarios. From the reconstructed results shown in Fig. 4, it can be found that the reconstruction results of the X-Space method are poor due to the influence of the radial PSF, and there is an obvious gap with the ideal concentration distribution of magnetic particles (ground truth). However, PMCNet avoided this problem and the reconstruction results were very close to the ground truth. Especially for the letter phantom, the reconstructed result was exactly the same as the ground truth. However, as can be seen from the vessel and concentration phantoms, the results of our PMCNet method become blurred at the edges of the phantoms due to the complexity of the reconstructed scene. Nevertheless, the PMCNet method still has significant improvements compared to the X-Space method and SM method. To further evaluate the reconstruction results quantitatively, we calculated the indicators of PSNR and SSIM, as listed in TABLE I. Letter, Concentration and Vessel respectively represent three different phantoms. We find that the PMCNet framework significantly improved PSNR and SSIM compared to the X-Space method.

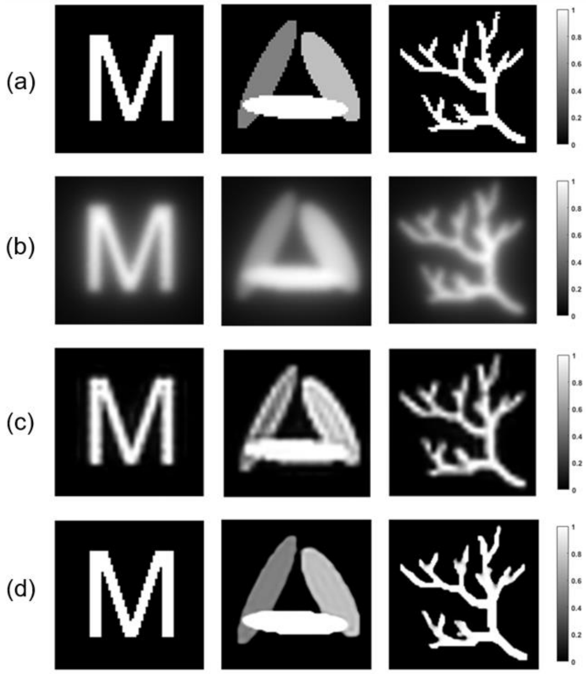

Fig. 4. Reconstruction results of accuracy evaluation experiments with three designed phantoms in complex scenarios. (a) Ground truth of two-dimensional phantoms, including the letter phantom, concentration phantom, and vessel phantom. (b) The reconstruction results of the X-Space method. (c) The reconstruction results of the SM method. (d) The reconstruction results of the PMCNet framework.

TABLE I

EVALUATION INDICATORS OF PSNR AND SSIM FOR ACCURACY EVALUATION EXPERIMENTS

		Letter	Concentration	Vessel
PSNR	X-Space	11.775	14.094	10.222
	SM	17.423	19.834	14.919
	PMCNet	25.950	25.763	20.047
SSIM	X-Space	0.083	0.206	0.060
	SM	0.639	0.656	0.585
	PMCNet	0.979	0.933	0.934

B. Spatial resolution evaluation

To further investigate the spatial resolution of the proposed PMCNet framework, we designed line-pair phantoms with an increasing spacing from 0.2 mm to 0.8 mm. The ground truth as well as the reconstruction results by different methods are presented in Fig. 5. Similar to the results of accuracy evaluation experiments, the X-Space method can only provide limited spatial resolution due to the radial PSF. Reconstructions of line pairs at 0.8 mm are able to be barely distinguishable. However, the results of the PMCNet framework are not affected by PSF artifacts. The reconstruction results for line pairs are extremely well differentiated in all the investigated cases. To define the spatial resolution more intuitively, we extracted a line through the center of the line-pair phantom and then plotted the intensity profiles of the reconstructed results along the extracted line, as shown in Fig. 5(d). As can be seen from the results, since it does not require the use of PSF, the PMCNet framework is able to achieve a spatial resolution of better than 0.2 mm, which is more than four times better than the results of X-space method and SM reconstruction.

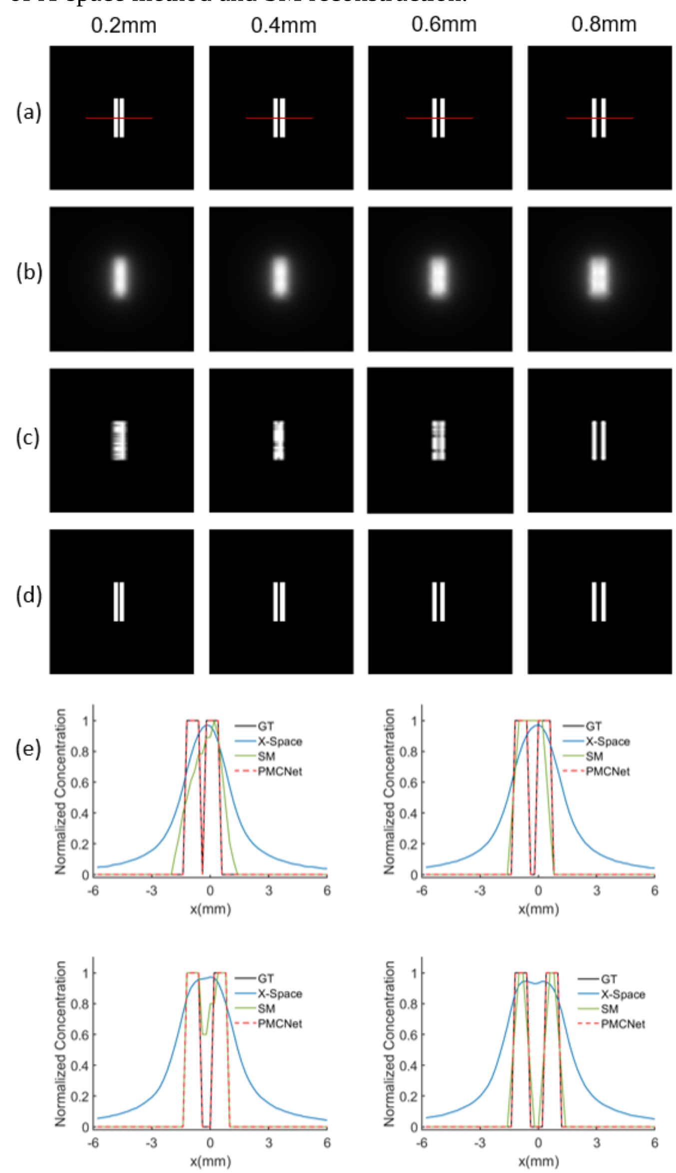

Fig. 5. Reconstruction results of spatial resolution evaluation experiments. (a) Line-pair phantoms with different spacings of 0.2, 0.4, 0.6, and, 0.8 mm. (b) The reconstruction results of the X-Space method. (c) The reconstruction results of the SM method. (d) The reconstruction results of PMCNet. (e)

Intensity profiles of the reconstructed results on a line extracted through the center of the line-pair phantoms (indicated by red lines in (a), at y=0 mm).

C. *Noise immunity evaluation*

To explore the noise immunity of the PMCNet framework, we next added different levels of Gaussian white noise to the simulated data to more realistically simulate the actual scanning process. We compared the robustness of the PMCNet and X-space methods at high noise levels. 15dB, 20dB, and 25dB of Gaussian white noise were added to the data, respectively. It should be noted that we did not perform any denoising on the voltage signal before reconstruction. The X-space method is sensitive to noise, especially in regions with low scanning speed, resulting in only noise in the normalized reconstruction results. Therefore, we only present the reconstruction results of the PMCnet here, as shown in Fig. 6. We can find that PMCNet is able to reconstruct the approximate distribution of MNPs (Fig. 6(c-d)), although many noise points appear in the reconstruction results at high noise level (Fig. 6(b)). Moreover, PMCNet can reconstruct satisfactorily when the signal-to-noise ratio is 20 or higher.

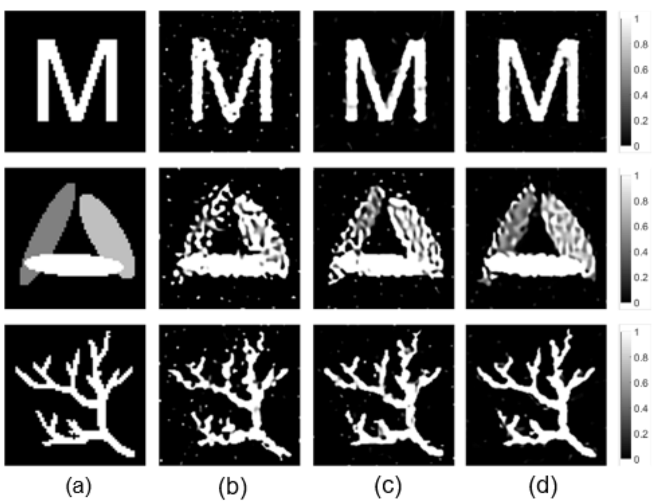

Fig. 6. Reconstruction results of noise immunity evaluation experiments by PMCNet. (a) ground truth of two-dimensional phantoms, including letter phantom, concentration phantom, and vessel phantom. (b)-(d) reconstruction results using noise-added data with the signal-to-noise ratios of 15dB, 20dB, and 25dB respectively.

TABLE II
EVALUATION INDICATORS OF PSNR AND SSIM FOR NOISE IMMUNITY EVALUATION EXPERIMENTS

		Letter	Concentration	Vessel
PSNR	15dB	16.422	12.767	12.779
	20dB	19.283	14.216	14.951
	25dB	19.681	17.149	16.083
SSIM	15dB	0.681	0.500	0.589
	20dB	0.829	0.548	0.721
	25dB	0.839	0.635	0.784

We also calculated the PSNR and SSIM to quantitatively evaluate the noise immunity performance, as detailed in TABLE II. In terms of quantitative metrics, we find that the addition of noise has a large impact on the reconstructed image quality. As the signal-to-noise ratio of the data increases, both the PSNR and SSIM are gradually improved. However, even with a signal -to-ratio of 25 dB, the PSNR and SSIM still fall short of the reconstructed results in the ideal case.

D. *Non-adiabatic data reconstruction*

To better match the actual process of MPI, we incorporated Debye relaxation into the physical model to simulate the acquisition of nonadiabatic data. We used the gradient descent algorithm to estimate the relaxation time constant of particles while reconstructing the spatial distribution of MNPs. The results are shown in Fig. 7. Under such simulation conditions, there is a slight error in the estimation of the concentration of MNPs at phantom boundary by the PMCNet framework, which is mainly reflected in the presence of reconstruction artifacts in the concentration phantom and the vascular phantom. These artifacts in the reconstruction results become more obvious as the relaxation time constant increases. However, the results of the PMCnet framework for nonadiabatic data with relaxation effects are still within acceptable limits. In particular, the estimation of the relaxation time constant by PMCNet is very accurate with an estimation error of less than 1%, as shown in Table III.

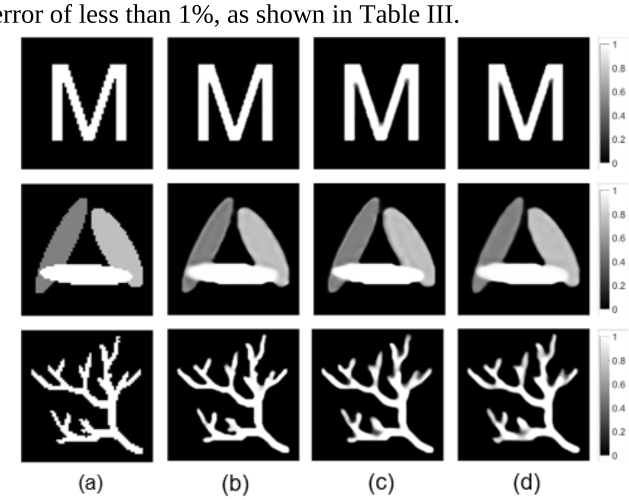

Fig. 7. Non-adiabatic data reconstruction results. (a) ground truth of two-dimensional phantoms, including letter phantom, concentration phantom, and vessel phantom. (b)-(d) reconstruction results using nonadiabatic data with relaxation time constants of 1 μs, 2 μs, and 3 μs, respectively.

TABLE III
ESTIMATION RESULTS OF THE RELAXATION TIME CONSTANT

GT	1 μs	2 μs	3 μs
Letter	0.999	1.998	2.997
Concentration	0.999	1.998	3.001
Vessel	0.999	1.999	3.005

E. Multi-color MPI reconstruction

Finally, we arrived at the concept of employing PMCNet to achieve multi-color MPI reconstruction. To prove this concept, we designed a letter phantom that consists of a letter "X" and a letter "D", as described in Fig. 8(a). These two letters represent two types of MNPs with varying relaxation time constants. The relaxation time constant of the letter "D" is set to 2 μs and 5 μs for two groups of investigation, and that of the letter "X" is set as 1 μs. Here multi-color refers to different values of relaxation time constants. Multi-color MPI reconstruction is to reconstruct the distribution of MNPs with varying relaxation time constants by MPI, while different MNPs can be displayed with different channels.

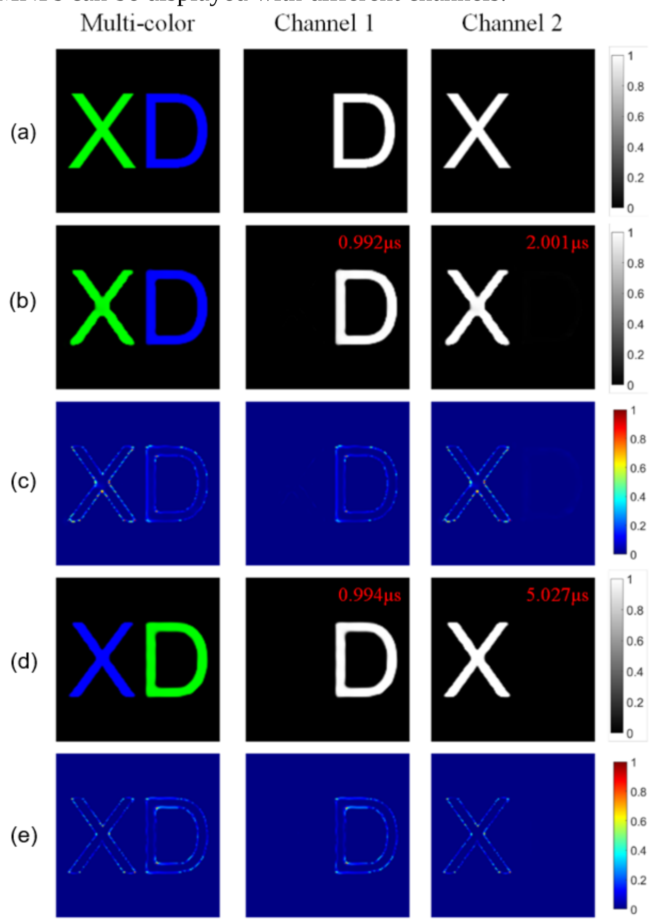

Fig. 8. Multi-color MPI reconstruction results. (a) The letter phantom that consists of a letter "X" and a letter "D", representing two types of MNPs. (b) and (d) represent the reconstruction results of multi-color MPI. Here, the relaxation time constant of the letter "D" is set to 2 μs and 5 μs, and that of the letter "X" is set to1 μs. The red numbers represent the estimated value of the relaxation time constants. (c) and (e) are error maps between the reconstructed results and the ground truth.

Figures 8(b)-(e) show multi-color MPI reconstruction results. From these results, it can be seen that PMCNet can realize intact individual reconstruction of MNPs with different relaxation time constants and accurately estimate the values of the relevant relaxation time constants with an error of less than 1%. Similar to non-adiabatic data reconstruction, the errors in the multi-color MPI reconstruction results are mainly concentrated at the edges of the phantom. To illustrate this more intuitively, we differ the multicolor MPI reconstruction result from the ground truth, and the difference maps are shown in Fig. 8(c) and 8(e). We find that the difference between them is concentrated at the phantom edges. More importantly, the PMCNet framework is able to reconstruct the distribution of MNPs as well as accurately estimate their relaxation time constants, regardless of whether the difference in the relaxation time constants of MNPs is large or small. From the results in Figs. 8(b) and 8(d), it can be seen that when the relaxation time constants of the two MNPs are similar, PMCNet can also estimate the distribution and types of particles correctly, although some artifacts appear in Fig. 8(c). These artifacts can become less or even eliminated as the difference in the relaxation time constants of the two MNPs becomes larger, as shown in Fig. 8(e)

V. Discussion

System matrix reconstruction requires significant time for measurement and calibration. While the X-Space reconstruction algorithm eliminates the need for a system matrix, it still depends on the accuracy of the Point Spread Function (PSF) and is susceptible to artifacts that are difficult to remove. Deep learning methods, on the other hand, primarily focus on system matrix recovery or system-specific training, which requires large amounts of training data. In this study, we propose PMCNet, an unsupervised MPI reconstruction framework that directly reconstructs the spatial distribution of MNPs from voltage data without relying on a system matrix. Compared to the X-Space method, PMCNet eliminates reconstruction artifacts and significantly improves the spatial resolution of MPI. Additionally, PMCNet utilizes physical model constraints to update the neural network parameters, avoiding the need for large training datasets.

Furthermore, PMCNet is robust to noise and can directly reconstruct noisy data without the need for separate denoising procedures. We have incorporated the Debye relaxation model into the physical model, and employed a gradient descent algorithm to estimate the relaxation time constant during reconstruction. The results demonstrate that PMCNet accurately reconstructs the spatial distribution of MNPs, even in the presence of relaxation effects, and estimates the relaxation time constant with less than 1% error. By increasing the number of channels, PMCNet can simultaneously reconstruct MNPs' spatial distribution and estimate their relaxation time constants, enabling multi-color MPI reconstruction. Simulation experiments highlight the method's potential for improved multi-color MPI applications.

Despite these advantages, there are still areas for improvement. In terms of reconstruction efficiency, although PMCNet does not require training, it still needs several iterations to optimize the network parameters. In our study, we used 20,000 iterations to ensure stable results, which significantly impacts reconstruction speed. While the inference time of PMCNet is comparable to existing iterative methods, its primary advantage lies in the bypass of the laborious system matrix calibration, thereby significantly streamlining the imaging workflow. Furthermore, while the physical model used here incorporates the Debye relaxation model, it is not the most accurate representation of magnetization behavior in real-world MPI scans. To illustrate this issue, we performed a real-world reconstruction using the Open MPI Data, which is a public dataset constructed by University Medical Center Hamburg-Eppendorf [25]. The results reconstructed using the PMCNet was compared with those by the measured system matrix provided in Open MPI Data, as shown in Fig. 9. Among them, the first row shows the reconstructed results using the calibrated system matrix, and the second row is the results generated by PMCNet. The system matrix was obtained via scanning under the 2D Lissajous trajectory with a grid size of 37×37×37. The middlemost layer was selected for demonstration.

It can be observed that the reconstruction performance of PMCNet is not entirely satisfactory on Open MPI data. For the resolution phantom, only regions with strong signals can be reconstructed well, while some shape features are lost. As for the phantom with varying concentrations, only the regions with the high concentration can be reconstructed. This is attributed to the physical model's deviation from real-world MPI scanning conditions, which prevents the network from converging effectively. In other words, during the iterative process of PMCNet, regions with strong MPNs signals can be approximated relatively well, whereas regions with weak signals require precise physical model.

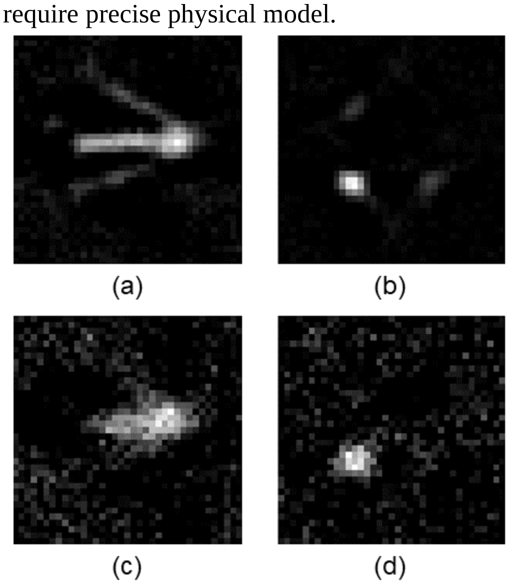

Fig. 9. Reconstructed results of Open MPI data by using the system matrix-based method and the PMCNet. (a)-(b) Results of the resolution and concentration phantoms reconstructed by the system matrix-based method; (c)-(d) Results obtained by the PMCNet.

Future work could address the following areas for improvement:

1. Noise Constraints: Although the current framework demonstrates robustness to noise, the present study considers only Gaussian noise. In practical MPI systems, noise characteristics are often more complex, involving frequency-dependent, system-induced, and non-Gaussian components. Future work will focus on extending the model to handle such realistic noise scenarios more effectively.

2. Relaxation Model: In order to quickly verify the effectiveness and feasibility of our proposed method, we used a simple relaxation model. In the subsequent work, we will add more complex and realistic models for comparative studies to further validate the effectiveness of our method.

3. Optimizing Network Structure: We will explore different neural network architectures to better adapt to the varied complexities of MPI data and improve both the speed and accuracy of the reconstruction.

By addressing these limitations, we aim to improve PMCNet's efficiency and accuracy, bringing it closer to practical applications in MPI.

VI. CONSLUSION

We propose an MPI reconstruction method called the PMCNet framework, which can reconstruct the spatial distribution of MNPs directly from time-series voltage data without the need for a system matrix and network training. Compared with existing analytical methods that do not require system matrix, PMCNet is able to suppress artifacts in the reconstruction results and significantly improve reconstruction quality and spatial resolution. Compared with deep learning methods that require large amount of training data, PMCNet obtains reconstruction results without training data due to the introduction of physical model constraints. The framework also has superior noise robustness, while the relaxation time constants of different types of MNPs can be estimated during the reconstruction process. Simulation experimental results show that PMCNet can accurately estimate the relaxation time constants of MNPs, thus demonstrating its potential application in multi-color MPI.

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